

GEOMATRIX — Roadmap & Actionable Next Tasks

Smart India Hackathon 2026 · Problem Statement 26017 · Land Acquisition Risk Intelligence

TASK 1: Initialize Database & Load Baseline Data

[] **1.1 Seed SQLite Database** — Run the database seed script to populate historical delay records and initial infrastructure projects:

Command: `cd geomatrix_v2 && python seed_db.py`

Output: Creates geomatrix.db with 20 historical training records & 6 baseline projects.

[] **1.2 Verify FastAPI Environment Variables** — Check `.env.local` contains `GOOGLE_GEMINI_API_KEY` and `NEXT_PUBLIC_API_URL='http://127.0.0.1:8000'`.

TASK 2: Start Backend API & Train ML Model on Real Data

[] **2.1 Launch FastAPI Server** — Start backend server:

Command: `cd geomatrix_v2 && python -m uvicorn main:app --reload --port 8000`

Verify: `http://127.0.0.1:8000/health` returns `{'status':'ok'}`.

[] **2.2 Ingest Real MoSPI CSV Dataset** — Navigate to /data page (Data Ingestion) and upload your MoSPI PAIMANA Flash Report CSV under 'Historical Data' (`data_type=historical`).

[] **2.3 Train ML Risk Classifier** — Trigger training via /model page or POST /api/model/train. Model will save RandomForest/XGBoost artifacts and output Precision, Recall, F1, and ROC-AUC scores.

TASK 3: Ingest Active Projects & Run AI Risk Predictions

[] **3.1 Ingest Active Projects CSV** — Upload MoSPI/Project CSV under 'Projects Data' (`data_type=projects`) via Data Ingestion screen (/data).

[] **3.2 Run Risk Prediction & SHAP Explainability** — Open any project in /projects/[id], click 'Run Risk Prediction'. Verify Risk Score, Risk Level Badge (Critical/High/Medium/Low), and SHAP Feature Importance chart.

[] **3.3 Test What-If Simulator** — Adjust legal dispute and pending claim sliders on project detail page to observe live risk score changes.

[] **3.4 Generate Gemini Mitigation Insights** — Click 'Generate AI Mitigation Plan' to test Gemini LLM decision recommendations.

TASK 4: Verify GIS Map, Early Warnings & Reporting Modules

[] **4.1 Verify GIS Spatial Map (/map)** — Confirm project GPS coordinates display colored risk markers with popups and district filters.

[] **4.2 Verify Early Warning Alerts Hub (/alerts)** — Confirm automatic alert generation for delay bottlenecks and test Acknowledge/Resolve actions.

[] **4.3 Verify PDF & Excel Export (/reports)** — Export project risk summary reports to PDF, Excel (.xls), and CSV.

TASK 5: Final Production Build & Hackathon Presentation Sequence

[] **5.1 Run Production Build Check** — Verify Next.js compiles without errors:

Command: `npm run build`

[] **5.2 Dry-Run Hackathon Presentation Sequence** — Order: Demo Login → Executive Dashboard → Live MoSPI CSV Ingestion → Train ML Model Live → Predict & SHAP Deep-Dive → What-If Simulation → Executive PDF Export.

Quick Execution Steps Summary

Sequential Task Step	Execution Command	Directory
1. Seed Database	<code>python seed_db.py</code>	geomatrix_v2/
2. Run Pytest Suite	<code>python -m pytest tests/ -v</code>	geomatrix_v2/
3. Start FastAPI Server	<code>python -m uvicorn main:app --reload --port 8000</code>	geomatrix_v2/
4. Start Next.js Frontend	<code>npm run dev</code>	Root folder
5. Production Build	<code>npm run build</code>	Root folder